

Project Management Report

Distributed Social Media Backend

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Overview

The team approached the final project collaboratively, structuring work around key phases including design, implementation, and evaluation. Early on, we collectively broke down the system into core components, which guided the microservices architecture and ensured consistency across development. Responsibilities were distributed to enable parallel progress, with each member leading one experiment while supporting others throughout implementation and analysis. Frequent communication and joint debugging allowed us to iteratively refine the system, address integration and deployment challenges, and successfully execute all experiments. All team members also contributed to final deliverables, ensuring a cohesive and well-integrated outcome.

Challenges Encountered

Our team faced three primary challenges during this project: experiment design refinement, infrastructure complexity, and task dependency management.

Experiment design required multiple iterations to arrive at a testable and meaningful scope. Our initial proposal compared fan-out-on-write versus fan-out-on-read architectures, but through discussions with our teaching assistant and individual research, we refined Experiment 1 to compare Redis-cached timelines versus direct PostgreSQL reads within the same fan-out-on-write architecture. This refinement made the experiment more focused and realistic. Rather than testing two fundamentally different architectures, we isolated the specific impact of caching infrastructure on read-heavy workloads.

Infrastructure design consumed significant time, particularly Terraform configuration for AWS resources. Coordinating ECS services, PostgreSQL with read replicas, ElastiCache Redis, and networking between components required careful debugging and iterative refinement.

Task dependencies created parallelization challenges because foundational components had to be completed before dependent work could begin. For example, timeline testing could not proceed until the Tweet service was functional, and experiments required both the data layer and seed data to be ready. We resolved this by dividing work strategically: Susie focused on Terraform infrastructure, Elly began

testing immediately to validate infrastructure reliability as components became available, and Sharon created seed data for experiments in parallel. This approach allowed us to maximize productivity despite sequential dependencies.

Project Task Breakdown and Timeline

Week of 3/9:

- ☐ Finalize project proposal
- ☐ Team discussion & system design

Week of 3/16:

- ☐ Core service skeleton: initial microservices structure
- ☐ Infrastructure: AWS resources

Week of 3/23: Minimal working system + baseline testing

- ☐ Implement at least 1 end-to-end flow
- ☐ Schema design, write DDL for database
- ☐ Deploy services via Terraform
- ☐ Set up basic load testing, collect baseline metrics
- ☐ Project milestone 1 report + elevator pitch video

Week of 3/30: Core functionality + data layer completion

- ☐ All core functionality: User auth, tweet CRUD, follow/unfollow, timeline
- ☐ Integrate PostgreSQL (single instance => replicas)
- ☐ Integrate Redis
- ☐ Validate system under moderate load
- ☐ Prepare for experiment toggles (Redis on/off, consistency modes)

Week of 4/13: Experiments + finalization

- ☐ Experiment 1: Redis caching impact
- ☐ Experiment 2: Eventual vs. strong consistency
- ☐ Experiment 3: Failure resilience
- ☐ Run stress tests at scale (large concurrent users)
- ☐ Collect, organize, and analyze results
- ☐ Complete analysis of results, refine charts and findings
- ☐ Final deliverables (reports, codebase, video presentation)

Individual Contribution

Elly

- Wrote the detailed project proposal.
- Created initial structured codebase skeleton.
- Debugged and validated the system's deployment on AWS, ensuring that services could run successfully in the cloud environment.
- Contributed to the writing of the milestone 1 report.
- Responsible for experiment 3: completed tests and analysis.
- Contributed to the writing and creation of final deliverables.

Susie

- Drafted the initial project proposal, created elevator pitch video.
- Conducted a detailed walkthrough of the generated codebase, made necessary modifications to align the implementation with system requirements.
- Developed Terraform configuration.
- Drafted milestone 1 report and README for the codebase.
- Responsible for experiment 1: completed tests and analysis.
- Contributed to the writing and creation of final deliverables.

Sharon

- Completed the Methodology and Preliminary Results sections of the milestone 1 report.
- Validated the database layer setup, conducted system testing in the cloud environment.
- Created seed data scripts, executed baseline load tests to collect baseline performance data.
- Responsible for experiment 2: completed tests and analysis.
- Contributed to the writing and creation of final deliverables.